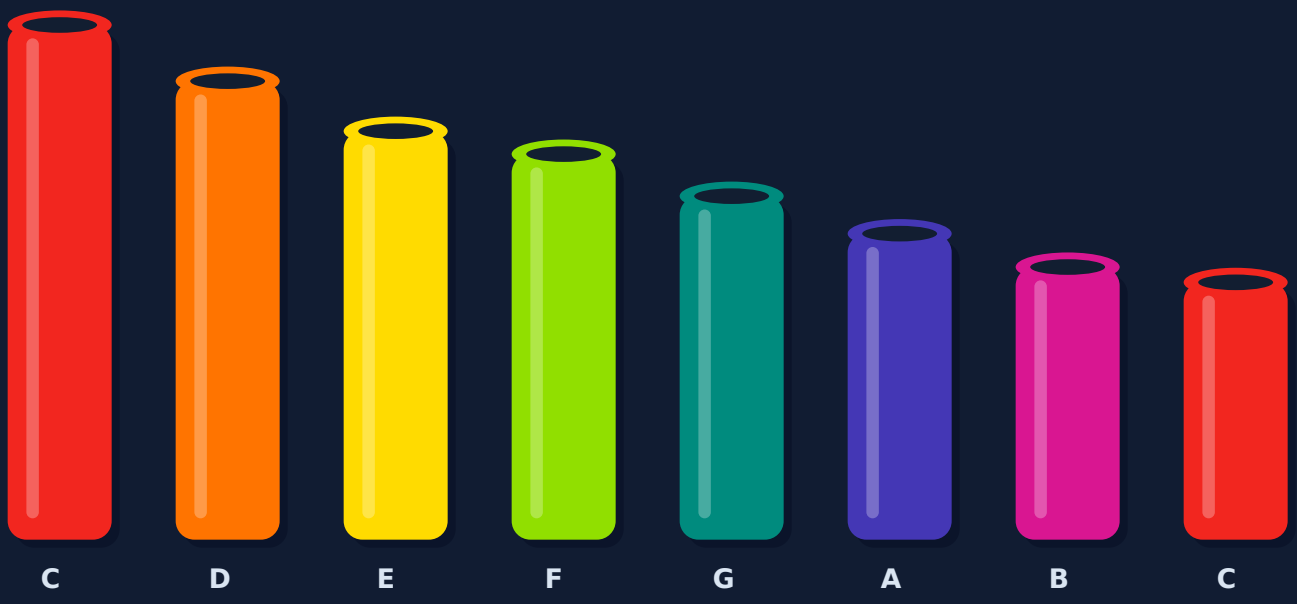
MR. DREW’S MUSIC LAB

TEACHER EDITION

Boomwhacker Play-Along

The complete user guide

Compose. Rehearse. Perform.



Created by Drew Wichman

Web app manual • October 9, 2026

A practical reference for teaching, composing, and playing together.

Find what you need

Read pages 3-5 first, or jump straight to the task you are working on.

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Two ways to make music

Use **Sections & measures** when you want reusable patterns and a musical grid. Use **Record Along** when you want to tap directly with a recording, without knowing its tempo.

Page numbers are clickable. PDF bookmarks offer another way to navigate. This guide covers the Teacher Edition web app and its classroom music tools.

Your first play-along

Make a short arrangement, hear it, and put it in front of your class.

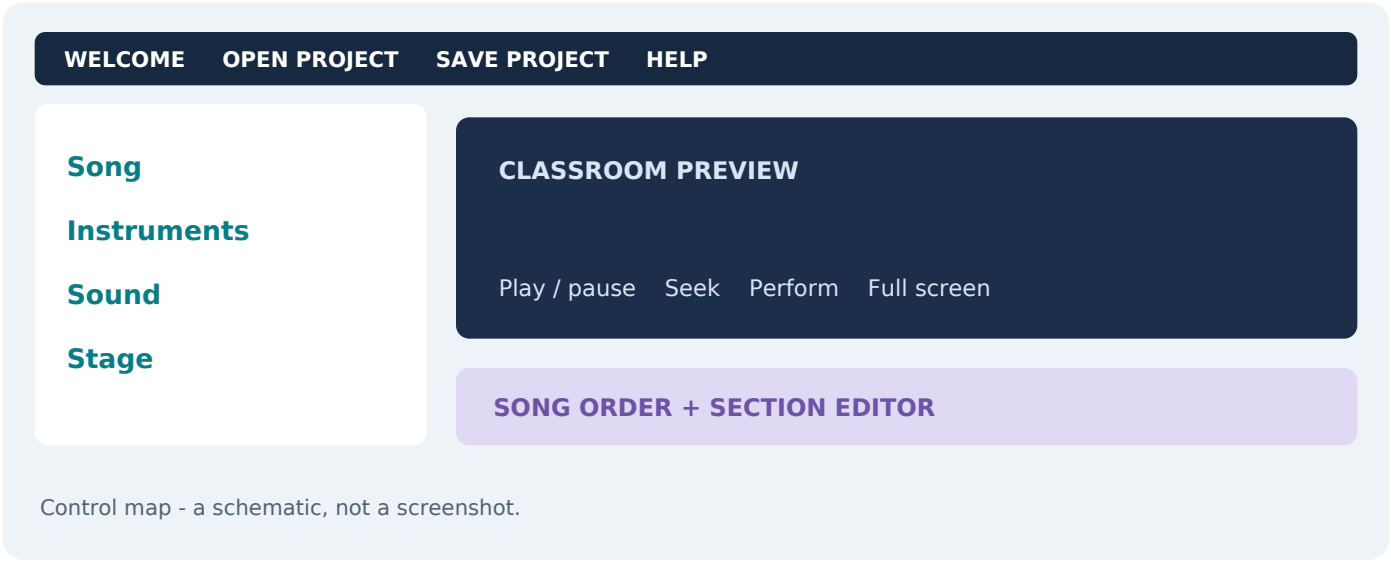
1. **Start creating.** Open the app and choose Start creating on the welcome screen. To hear an example first, choose Explore demo. Use Welcome in the toolbar to return to that screen.
2. **Name the song.** In Song, enter a title and Song tempo. For a first project, try 100 BPM. The title is also used when naming downloads.
3. **Choose your instruments.** In Instruments, keep Diatonic tubes or choose the parts your students will use. For a percussion-only activity, choose No Boomwhackers and turn on the percussion you need.
4. **Set up a section.** Beside the editor, choose 4/4, four Section measures, and an eighth-note Grid snap. You can use the existing section or create a named A Section.
5. **Add a simple pattern.** Click empty grid cells to place strikes. Put notes at the same time in different rows for a chord. Preview notes lets you hear the pitch as you enter it.
6. **Try the whole song.** Check Song order and add your section if necessary. Choose Full song and press Play. In Stage, choose your view, count-in, introduction, and background.
7. **Save, then perform.** Click Save project and keep the downloaded file. Choose Perform for the classroom display; use Full screen if desired. Escape returns from performance mode.

> ### A small first pattern
>
> In a 4/4 section, try C on beat 1, E on beat 2, G on beat 3, and C on beat 4. Repeat or vary it for four measures. Add a clap on beats 2 and 4 in the falling-note views.

Start with a few lanes so students can see their part easily. Add the rest of your arrangement after the basic rhythm feels clear.

Find the controls

The four settings tabs organize most of the app. Section controls stay beside the note editor.



Control map - a schematic, not a screenshot.

Where	What you will find
Song	Title, tempo, Load demo, New arrangement.
Instruments	Tube sets and octaves, bass equipment, mallet parts, singing melody, percussion, instrument order and percussion sound.
Sound	Mixer, backing audio, audio alignment, MIDI import, Boomwhacker sound, guide notes, metronome.
Stage	Play view, scenery, introductions, bar setup, labels, loops, endings and visual timing.
Beside / below the editor	Song order, sections, meter, grid snap, precise entry, rolls, lyrics.

On a laptop: Hide settings expands the editor and play area. Show settings brings them back. Stage settings opens that tab directly. Small windows may begin with settings tucked away. Scroll inside long dialogs, the grid, song cards, or bar racks to reach their remaining controls.

Save, open & older projects

Your project file is the editable master. Save it to keep and reuse your arrangement.

Save a working copy

Click **Save project** in the top toolbar. The app downloads a file ending in **.bwstudio.json**. It contains your sections, song order, notes, lyrics, melody, instrument choices, mixer, visual settings, recorded strikes, and imported backing audio.

> ### Save deliberately
>
> The app does **not** automatically save your arrangement. Save before closing the tab, loading another project, starting a new arrangement, or making a major revision. A browser warning about unsaved work is not a backup.

Open and continue

1. Click **Open project** in the toolbar or welcome screen.
2. Choose your saved .bwstudio.json, .json, or .bwstudio file. Review any replacement confirmation before continuing.
3. Press Play briefly and check the instrument selection and song order. Save a new copy after your edits.

File / setting	What to expect
Older single-pattern save	Opens as a reusable section rather than losing its original music.
Older section-based save	Sections and their timing are retained. New saves use the mixed-meter format and should be reopened in this version or later.
Older Tubano / glockenspiel	The original Tubano maps to Medium; the original glockenspiel maps to Alto glockenspiel.
Browser preferences	Show at startup and editor note-preview preference are remembered by that browser. They are not a substitute for saving the song.

Keep dated copies, for example Spring_Concert_v03.bwstudio.json. The website loads its built-in artwork and sounds from the internet; the project file is not a standalone copy of the website.

Build a song with sections

A section holds the music. Song order tells the app when to play it.



Every A card refers to the same A pattern unless you make a copy.

1. Choose **New section**. Name it, select its time signature, and set its length. Leave Add this section to the end of the song checked if you want it in the arrangement immediately.
2. Select the section in **Edit section** or click its song card. Enter its notes, rolls, melody, and lyrics.
3. Use **Add to song** to place an existing section again. Set the repeat count on a song card, or add another occurrence later in the form.
4. Drag song cards to reorder them, or use their left/right arrows. The cards show their meter and position in the song.

Play command	Scope
Preview section / Selected section	Plays the section currently being edited.
Play song / Full song	Follows Song order from beginning to end.
Play button on a song card	Starts the full-song timeline from that entry; rehearsal looping is switched off.

A section can contain **1-512 measures**; a song entry can repeat it **1-16 times**. Editing a reused section updates every occurrence, including its melody and lyrics. Use an independent copy for a variation or a different verse.

Backing audio stays in place

Rearranging sections moves the note parts, not the recording. If you are following backing audio, recheck the timing after changing the form. Section preview uses that section’s first occurrence in the song.

Copy, remove & delete

Choose the action that matches how much of the arrangement you intend to change.

Action	Result
Copy section above the editor	Creates an independent copy of the selected pattern and adds it to Song order once.
Copy section on a song card	Inserts an independent copy after that card and retains its repeat count.
More actions > Rename	Changes the section name everywhere that same section is used.
Repeat page	Copies the visible editor page, up to four measures, immediately after that page.
More actions > Make page a section	Copies the visible measures into a new reusable section. The original stays intact.
Remove on a song card	Removes only that occurrence from Song order. The underlying section stays available.
Delete section everywhere...	Deletes the section, its notes, rolls, melody, and lyrics, and all of its song entries. A confirmation explains what will be removed.

A safe way to make a new verse

1. Select the original section and choose **Copy section**.
2. Rename the copy, for example A - verse 2.
3. Change its words or notes. The original section remains independent.
4. Move the new song card to the correct place and play across both transitions.

Undo restores section copies, deletions, song-order changes, and percussion reordering during the editing session. Use Redo to reapply an undone edit. These controls do not replace a saved project backup.

At least one section must remain. To empty the last section, clear its notes instead of deleting the section. After changing song length or meter, review rehearsal ranges and any audio-alignment markers.

Time signatures & tempo

Each section can have its own meter. Tempo is set in Song; timing follows the selected section’s pulse rules.

Meter	Numbering in the editor	Musical grouping
2/4	Beats 1-2	Two quarter-note beats.
3/4	Beats 1-3	Three quarter-note beats.
4/4	Beats 1-4	Four quarter-note beats.
6/8	Subdivisions 1-6	Two main pulses: 1-2-3 and 4-5-6.



6/8 count-in: STRONG weak weak STRONG weak weak. The grid reflects the same two groups.

Choose what 6/8 does at a meter change

Keep main pulse makes the two dotted-quarter beats follow Song tempo. At 100 BPM, each big beat lasts 0.6 seconds, and a 6/8 measure lasts 1.2 seconds.

Keep note values preserves quarter-note speed. At 100 BPM, each eighth note lasts 0.3 seconds, and the 6/8 measure lasts 1.8 seconds. Read the section’s pulse hint to see the resulting dotted-quarter tempo.

Grid snap and existing music

Grid snap offers quarter, eighth, or sixteenth notes. It controls note entry and dragging. Lyrics have a separate timing step. Changing meter preserves existing note positions and re-bars a populated section; inspect its new measure count and phrase boundaries afterward.

Song tempo accepts 30-240 BPM. With active audio alignment, marker spacing controls the timing and the 6/8 pulse selector is disabled. Sections join continuously; the opening count-in is not inserted between every section.

Enter and edit notes

One block is one strike. Simultaneous blocks create a chord or ensemble hit.

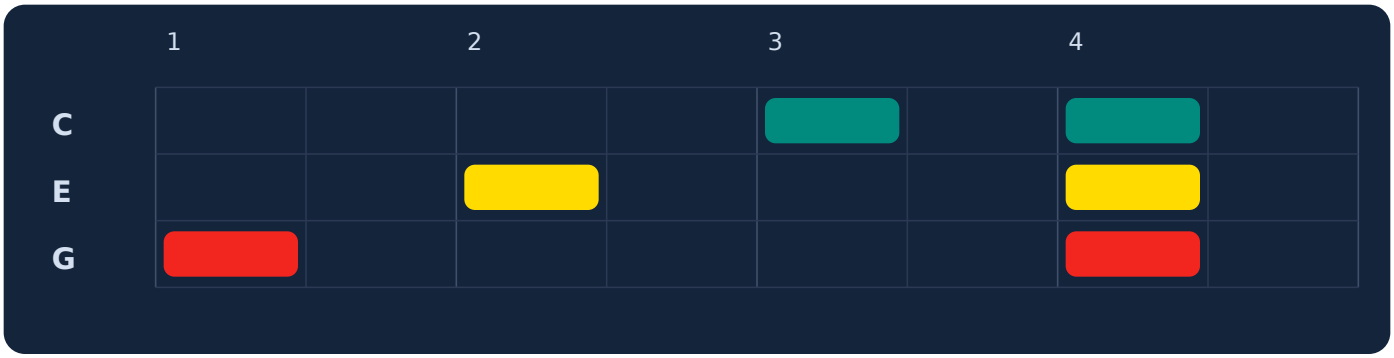


Illustration: one 4/4 measure at eighth-note snap, ending with a three-note chord.

Gesture / control	What it does
Click an empty cell	Adds a strike at the snapped time and instrument row.
Click an existing note	Selects and auditions it; fills the precise-entry controls.
Drag the note body	Moves the strike in time or pitch.
Erase, right-click, Remove strike	Deletes a strike silently.
Drag the right-edge grip	Extends the note into a roll; see page 10.
Previous / next page	Moves through pages of up to four measures.
Clear page	Clears notes on the visible page; use Undo if needed.

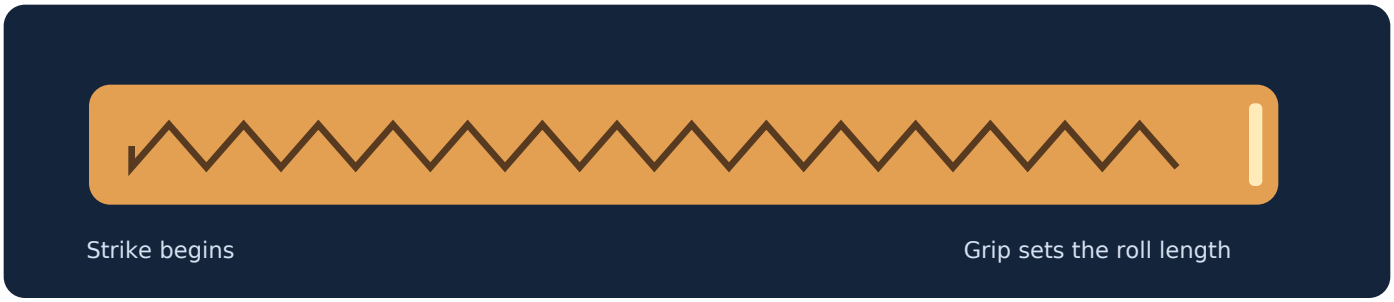
Precise note entry & measure editing gives you Instrument, Measure, Beat (or Subdivision in 6/8), and Play type. Choose Add / update to commit the position, or Remove strike to remove the exact strike there.

Hear what you are adding

Preview notes is on by default and can be switched off. It uses the selected sound, mixer level, bass substitution, and damper setting. Changing pitch auditions the new note; moving only in time or resizing does not replay it.

Rolls & repeated playing

Extend a strike when you want continuous rolling, shaking, or repeated tapping.



The zigzag distinguishes repeated playing from a single strike.

1. Add a note, then drag its small **right-edge grip** to extend it.
2. For exact editing, select the note and open **Precise note entry**. Choose Roll, enter its length, and choose Relaxed taps, Fast roll, or Very fast roll.
3. Click **Add / update**. Play the passage and listen to the result.
4. To return to one hit, choose **Single strike** at that position and click Add / update.

How rolls behave

- The guide sound uses fast, slightly uneven taps with small touch and tone differences. Roll speed is independent of the song tempo.
- A roll stops at its section boundary. Its entered length must fit inside the section.
- Rolls copy with sections and repeated pages. Dragging the note body moves the whole roll.
- In 6/8, the length fields use eighth-note subdivisions; in simple meters they use quarter-note beats.

> **For the final note of a song**
>
> Leave enough time inside the last section for the whole roll. An ending message is not extra musical space: extend the section if students need to keep rolling longer.

Try rolls on Tubanos, hand drums, tambourine, jingle bells, or pitched instruments. For maracas and güiro, a roll means repeated shakes or scrapes. The singing-melody editor uses a different behavior: stretching a melody note sustains one tone.

Boomwhackers & bass options

Start with the tubes you actually want students to use, then add other parts.

Control	Purpose
No Boomwhackers	Removes all tube lanes across the enabled octaves. Existing tube notes are retained but hidden and muted.
Diatonic / Pentatonic / Chromatic	Chooses a starting tube set. Tap individual pitch buttons to include or hide parts.
Standard range	C4-C5. This is the normal starting range.
Add lower octave	Adds C3-B3; choose the equipment in Bass setup.
Add higher octave	Adds C5-C6. With both extensions, the software covers C3-C6.

Four ways to cover the lower part

- **Standard tubes + Octavators:** the introduction identifies both the tube students hold and the lower sounding pitch, such as D4 + cap for D3.
- **Bass tubes:** use actual longer tubes for the low notes.
- **Bass xylophone:** the lower part uses a synthesized xylophone sound and a barred-instrument display.
- **Bass metallophone:** the lower part uses the metal-bar sound; the damper control is shared with that bass metallophone part.

Registers are distinguished by octave labels and a glowing red divider in the falling-note views. C5 appears once, in the higher group when that extension is enabled. The upper sample range uses pitch cards in the introduction; match those pitches to the equipment available in your classroom.

Recorded sound and accidentals

Sound > Boomwhacker sound offers Recorded Boomwhackers or Synthesized guide. The 22 natural-note recordings span C3-C6. C-sharp, E-flat, F-sharp, A-flat, and B-flat are made by lowering the next natural sample by one semitone. Hear C4 auditions the selected sound.

Switching off an octave or a pitch hides and mutes its part without erasing its saved notes. Selecting a tube set after No Boomwhackers brings tube parts back into use.

Barred instruments

Choose a family under Mallet instruments, then enable each part and select its grid pitches.

Part	Short name	Sounding range
Bass xylophone	BX	C3-A4
Alto xylophone	AX	C4-A5
Soprano xylophone	SX	C5-A6
Bass metallophone	BM	C3-A4
Alto metallophone	AM	C4-A5
Soprano metallophone	SM	C5-A6
Alto glockenspiel	AG	C5-A6
Soprano glockenspiel	SG	C6-A7
Contrabass bars	CB	Individual pitches within C2-C3

The complete xylophone, metallophone, and glockenspiel racks have **13 natural bars from C to A**, plus optional F-sharp and B-flat substitutions: two F-sharps and one B-flat in each range. Pitch numbers are sounding pitches, including on the glockenspiels.

Natural bars restores the normal set of grid pitches. **Clear** hides those grid pitches; saved notes remain. **Hear** auditions the instrument. Different instruments can play the same pitch as separate parts, each with its own mixer channel.

Sound and damping

Xylophones and contrabass bars use synthesized wooden-bar tones. Metallophones use sustained metal-bar tones; **Damper - short notes** shortens the sound, while switching it off lets notes ring. Glockenspiels have their own bright synthesized tone.

Individual bass bars

Contrabass choices are C2, D2, E2, F2, F-sharp2, G2, A2, B-flat2, B2, and C3. C / F / G restores the starter selection. Choose only the individual bars you have; these do not need a full-rack removal preview.

Choose bars to remove

Bar-removal tips are a visual teaching suggestion. They do not erase music or alter its sound.

1. Enable a barred instrument under Instruments and add its part to the song.
2. Open **Set up bars...** on that instrument, or use **Stage > Choose bars...**. Select the correct Instrument.
3. Set **Setup choice** to Choose bars myself. Click an unused bar to mark LIFT OFF; click again to put it back. Needed bars are protected.
4. Turn on **Show instrument introduction** and **Include bar-removal tips** in Stage. Use Preview screen to check what students will see.

Setup choice	Recommendation
Automatic from notes	Recognizes the complete supported pentatonic pitch sets; otherwise offers range markers.
Range markers	Suggests the nearest unused bar below and above the playing range, where available.
C major / A minor	Burgers and fries: remove B and F bars that are not needed.
F major / D minor	Eggs and bacon: remove E and B bars that are not needed.
Choose bars myself	Your custom lift-off choices for that instrument.
No removal tips	Hides this instrument’s removal advice. Find your bars can still show its positions.

Customize these suggestions starts a manual setup from the recommendation shown. **Keep all bars** clears removals. Undo restores the preceding edit. If a later score change needs a removed bar, the app protects that pitch and tells you to keep it available.

Why a bar will not lift off

A bar marked NEEDED is used in the song. Check every arranged section, not just the one currently visible. If a natural and its replacement accidental are both required, read the extra-bar notice and plan the physical setup accordingly.

Bar positions & unused gaps

Three separate controls serve three different teaching needs.

Control	Changes
Instruments > Pitches on the grid	Which pitches are selected as editable and playable parts.
Stage > Unused bars	Whether unused internal positions remain as dark, labeled gaps on the playboard. Each instrument has its own switch.
Set up bars / removal tips	What the introduction suggests lifting off the real instrument. Score and sound stay the same.



E is unused, but its dark position preserves the gap.

Keep useful gaps, hide unused ends

The playing range stays fixed across the song. Unused outer bars are omitted. With an instrument’s Unused bars switch on, unused natural positions **inside** that range remain dark and labeled. With it off, the board shows selected pitches within the range without dimming.

This helps students judge the distance between D and F even when E is not played. Contrabass parts retain only their chosen individual bars. A bass substitution and the matching independent bass instrument share the dimming switch.

Find your bars before playing

Turn on **Find your bars before playing** to show a complete instrument diagram with the song’s notes highlighted. First-C and next-C landmarks clarify octave position. Full instrument names remain below their lanes during performance; moving-note labels are separately optional.

Group introductions include each barred instrument’s diagram for at least six seconds. All together uses separate bar-preview pages with up to two instruments per page. These previews show the full rack even when unused ends are hidden during play.

Percussion & body percussion

Nineteen optional parts: fifteen instruments plus four body-percussion actions.

Group	Available tracks
Tubanos	Low, Medium, High - three separate recorded drums.
Hand drums	Low, Medium, High - three synthesized frame-drum sounds.
Metal / shaken	Triangle, Cowbell, Tambourine, Jingle bells.
Wood / scraped	Wood block, Vibraslap, Maracas, Güiro.
Large drum	Gathering drum.
Body percussion	Stamp, Pat, Clap, Snap.

Choose and arrange the parts

1. In **Instruments > Percussion & body**, use Find percussion, or the All, Body, and In use filters. Turn on each part you need. Click Hear to audition it.
2. Place notes in its labeled editor row. Its picture identifies the moving cue and target; body parts use action icons.
3. Turn on **Reorder**. Drag the handle or use arrow buttons, or drag a percussion name up/down in the grid. Top to bottom in the list becomes left to right on stage.

Reordering carries notes and rolls with the correct instrument and is saved with the project. When filtering, only visible parts move relative to one another; other tracks keep their relative order. Escape cancels a grid drag. Hidden tracks retain their notes and order.

Samples and guide sounds

Recorded samples cover the three Tubanos, triangle, wood block, cowbell, gathering drum, tambourine, jingle bells, vibraslap, maracas, güiro, clap, and snap. Jingle bells uses the sleigh-bell recording with the handheld bell-ring picture. Hand drums, stamp, and pat use synthesized sounds.

Percussion sound switches between Recorded samples and Synthesized guide. Play percussion sounds can be turned off while the cues remain for students. The overall percussion volume and individual Mixer channels both affect what you hear.

Balance the sound

Keep every visual part while adjusting how much of its guide sound is heard.

Control	Use it for
Boomwhackers fader	One shared level for the tube family.
Mallet faders	A separate level for each enabled barred instrument.
Percussion faders	Individual levels, listed in your percussion order.
Singing melody	Independent volume and Mute melody.
Backing volume	The loaded recording or built-in backing groove.
All percussion - overall volume	An additional master level for all percussion guide sounds, in Instruments.

A practical balancing routine

1. Play a busy passage and set the backing audio to a comfortable level.
2. Raise the Boomwhackers fader until pitches are easy to hear. **Boost tubes - 200%** is a quick starting point.
3. Lower loud percussion parts instead of raising every other part. Balance barred instruments and the singing guide last.
4. Save the project to keep this mix for your next classroom performance.

Mixer channels run from **0% to 400%**. At 100%, the original level is unchanged. At 0%, the sound is muted while its notes remain visible. Reset mix restores channel levels and the melody mix to their defaults.

> ### Muting is different from hiding
>
> To let students supply a part while still seeing cues, turn its mixer channel down to 0%, or use the guide-sound checkbox for that family. Hiding a tube pitch or disabling an instrument also removes its visible part.

Audition and metronome

Use Hear on an instrument, Hear C4 for tubes, or Hear sound for the melody. Sound > Metronome adds a timing guide. Set the count-in separately in Stage.

Add a singing melody

An audio-only tune can support singers while the visible board carries the accompaniment.

1. In Instruments, turn on **Play singing melody**. Choose **Doo - soft wordless voice** or **Gentle flute**, and use Hear sound.
2. Choose **Edit melody...**. The section selector follows the arranged song from beginning to end, including repeats; unused sections appear separately.
3. Click to add a note, drag its body to change pitch or time, and drag its right edge to sustain it. Leave gaps for rests.
4. For exact work, choose a Selected note and edit Pitch, Measure, Beat, and Length. Use Add note or Update note. Play this section checks the result.

What follows the arrangement

Melody notes belong to sections. Reusing A reuses A’s melody; copying A gives you an independent variation. The melody follows section meter and active audio alignment. It is monophonic: overlapping notes replace the overlapping span rather than create a choir chord.

The editor covers C3-C6. In simple meters, lengths are quarter-note beats; in 6/8, they are numbered eighth-note subdivisions. Preview notes is shared with the main editor. Delete note removes a melody event silently; Hear pitch is available for an explicit audition.

You want to...	Use...
Balance or silence the tune	Sound > Mixer > Singing melody / Mute melody.
Display actual words	Lyrics. The Doo voice does not pronounce the lyric text.
Keep the board uncluttered	The melody is already audio only; it adds no lanes or intro instrument.
Hear a single section	Edit melody > Play this section.

Choose the appropriate play view

The singing melody is part of section-based 2D and 3D arrangements. Bounce Along is restricted to the Boomwhacker part and does not play the separate singing melody. Record Along keeps its recorded strikes separate from section-based music.

Time your lyrics

Give students time to read the words before they need to sing them.

1. Open **Lyrics** above the preview or below the editor. Check that you are editing the correct section, then paste one short phrase per line and click **Add lines**.
2. Choose **Tap to time all lines**. After the count-in, press Space or Next phrase as each phrase begins. Finish timing keeps completed cues; untimed lines remain hidden.
3. Fine-tune each phrase with Measure and Beat fields, the earlier/later buttons, or the draggable timeline markers.
4. Set **Show words** to an early-display value, then play across section changes in **Full song**.

Timing control	Meaning
Lyric timing step	1 beat, half a beat, or a quarter beat. Separate from note-grid snap.
Beat 1.5	Halfway between beats 1 and 2. In 6/8, the fields use the six numbered eighth-note subdivisions.
Show words	On the beat, or 0.25, 0.5, 0.75, 1, 1.5, or 2 seconds early.
Add clear cue	A blank cue that hides lyrics for an instrumental passage.
Word timing	Optional per-word timing for the highlight; otherwise the app spaces words evenly.

Lyrics can lead across a section boundary

If B starts with a phrase on beat 1 and Show words is set to 1 second early, Full song can display that phrase during the end of A. Keep the phrase in B. Test the transition in Full song; Selected section does not include the preceding section.

Lyrics repeat with their section. Use Copy section for another verse. Re-timing all lines replaces their previous cue times; Undo restores them. The up/down buttons change the order used when tapping phrases, while the time fields determine playback timing.

Word timing: open a phrase’s controls and enter increasing offsets after the phrase begins. Use quarter-note beats, or eighth-note subdivisions in 6/8. Keep offsets before the next phrase or section end. Use even spacing resets them.

Use short phrases that fit comfortably on one or two lines. Early display adds reading time; changing the cue’s beat changes its musical position. These are separate decisions.

Style your lyrics

Choose a readable look first, then adjust the details to suit your students and stage.

Look	Visual direction
Classroom Bold	Strong, prominent classroom lettering.
Storybook	A softer storybook presentation.
Concert Caption	A cleaner caption treatment.
Retro Neon	Glowing pink tube-light lettering with subtle flicker.
Paper Cutout	A layered paper-inspired treatment.
Sing-Along / Clean - legacy	Retained for older projects and familiar styling.

Font, colors & placement

- **Font:** Match the style, Rounded, Modern, Friendly print, Storybook serif, or Bold poster. Fonts depend on what is installed on the computer.
- **Text size:** Small, Medium, or Large. Use View on a phrase to inspect it in the style preview.
- **Placement:** Top, Middle, or Bottom. Middle and Bottom overlay the playboard, so check that they do not hide the play cue.
- **Highlight:** Match the style, None, Current word, or Progressive fill. Open a phrase’s Word timing controls for more precise highlighting.
- **Motion:** None, Gentle fade, or Word emphasis. For steady Retro Neon, choose None.
- **Finishing controls:** text color, highlight/glow color, outline thickness, shadow depth, and Preview the next line.

> ### A clear classroom starting point
>
> Try Classroom Bold, Large text, Top placement, a modest outline and shadow, and Show words 1 second early. Add current-word highlighting only if it helps students follow the phrase.

Show lyrics switches the entire lyric overlay on or off. It is independent of Show song info and both pitch-label controls. Preview the actual phrases in performance before sharing them with your class.

Use backing audio

Your recording keeps its original speed and pitch. Choose how the notes will follow it.

Your goal	Use this workflow
Compose at a steady, known tempo	Sections & measures + Song tempo + First downbeat in audio.
Compose to a recording that changes tempo	Sections & measures + Align to audio. Map musical positions to the recording.
Play the parts directly, without a tempo map	Record Along - exact timing. Record each tap in seconds.

Load and place the recording

1. In **Sound > Music & sound**, click Load audio. WAV and MP3 are common choices. Other listed formats depend on what the browser can decode.
2. Set **First downbeat in audio - seconds** to the recording time at which measure 1 begins. For example, enter 2.40 if the first downbeat is 2.4 seconds into the file.
3. In section-based work, set the song tempo and arrangement length, or create an audio-alignment map. Adjust Backing volume.
4. Play from the beginning and later in the song. A good first beat does not guarantee that the whole recording stays aligned.

> ### Offset versus tempo
>
> If every note is early or late by the same amount, check the first downbeat or timing offset. If the difference grows as the song plays, check the tempo or use Align to audio. Changing Song tempo does not time-stretch the audio.

What is saved

Imported audio is processed in the browser and is included in Save project. Loading or rearranging a section does not cut, duplicate, or rearrange the audio file. The section notes move along a fixed full-song recording timeline.

The demo includes a built-in drum groove. To remove an imported track, use the remove-audio button beside Load audio; check the sound status afterward.

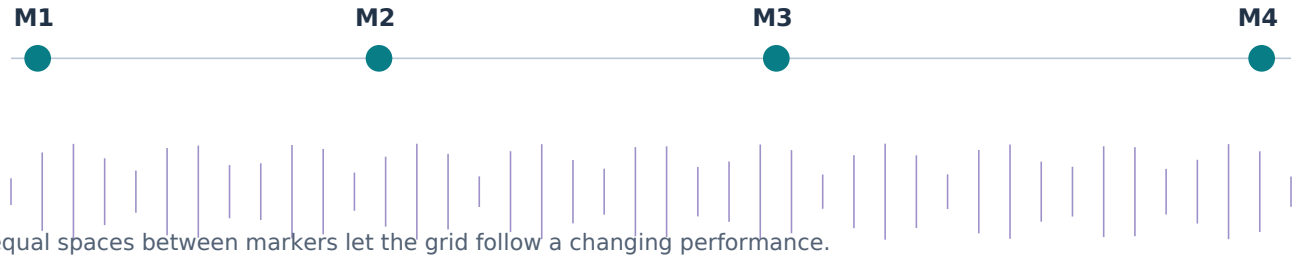
The audio loader is for a backing track. It is not a custom instrument-sample importer; the instrument recordings are built into this version of the app.

Align notes to a recording

Create a tempo map by listening and marking musical positions. The recording itself is unchanged.

1. Load the audio and build the song’s section order and meters. Open **Sound > Align to audio...**.
2. Listen or click the waveform to find measure 1. Click **Set first downbeat here**. Use Zoom, View, and Position in recording to navigate.
3. Choose **Tap every Measure** for a simple map, or **Main beat** for finer tempo changes. Set Starting at measure.
4. Click **Start tapping**, then tap Space or the Tap button at each requested position. Click Finish when done.
5. Choose a marker and refine its Measure, Beat, and Audio seconds, or drag it on the waveform. Add at playhead and Delete handle individual markers.
6. Keep **Follow this recording’s timing** checked and click **Apply timing**. Listen through transitions and the final measures.

MUSICAL POSITIONS



Unequal spaces between markers let the grid follow a changing performance.

At least two markers establish a pace. Timing between them follows their spacing; before the first and after the last, the nearest pace continues. Mark through the end of the song for the closest match. Preview metronome helps you hear whether the map agrees with the performance.

After structural changes

Reordering or deleting sections, changing meter, or changing song length can invalidate the relationship between markers and music. Reopen Align to audio and review it. Cancel abandons dialog edits; Apply timing commits them.

Record Along

Tap with an audio file using computer keys, on-screen pads, or a MIDI keyboard. No tempo is required.

1. Set **Create with** to **Record Along - exact timing**. Load a WAV or MP3 recording.
2. Choose Body percussion - four pads, Boomwhackers - pitched keys, or a single percussion instrument. For tubes, choose a computer-keyboard octave.
3. Use **Overdub - add another part** for another layer. Set Start at and, optionally, Stop at in seconds.
4. Press **Record**. After a silent two-second preparation, play along. Press Stop when finished, then Listen to review the take.

Input	Mapping
Body: computer keys	A = Stamp; S = Pat; D = Clap; F = Snap.
Body: MIDI note numbers	36 = Stamp; 38 = Pat; 39 = Clap; 42 = Snap. These are labeled C2, D2, D-sharp2, F-sharp2 in the app.
Tubes: computer keys	A W S E D F T G Y H U J K = chromatic pitches from C through the next C.
Tubes: MIDI	Uses actual pitches C3-C6, MIDI 48-84. Simultaneous keys can record chords.
Single percussion part	A, S, D, F, its on-screen pad, or any MIDI key triggers that chosen part.

Connect MIDI keyboard requests browser access. Select the device or All connected inputs. Computer keys and pads work without MIDI. The selected computer octave does not transpose MIDI's actual pitches.

What this mode records

Record Along records strike events and their exact times, not microphone audio or held-note durations. It supports tubes, percussion, and body percussion in 2D or 3D. It keeps section-based music separately; it does not convert the takes into section notes, lyrics, or mallet parts.

Hear my taps controls input monitoring. Hear recorded parts lets you hear prior layers while recording. No Boomwhackers must be changed to a tube set before recording tube parts.

Edit recorded takes

Clean up individual hits or move a whole take without stretching the recording.

Control	What it does
Waveform click	Moves the playback position and recording start time.
Zoom / Timeline position	Magnifies and moves the visible part of the recording.
Drag a strike	Moves its exact time on the timeline.
Selected strike / Exact time	Lets you choose an event and enter its time in seconds.
Earlier / Later buttons	Move the selected strike by 10 milliseconds.
Delete strike	Removes the selected event.
Take + Move take	Shifts the chosen take, or all recorded strikes. Positive values move it later.
Undo last edit / take and Redo	Restore or reapply edits, recordings, and replaced passages.

Replace a passage

Choose **Replace chosen instrument in range**, then set Start at and Stop at. Record the replacement. When the take contains new strikes, the old strikes for that chosen instrument in that range are replaced. A take with no new hits leaves the earlier part unchanged.

Correct a consistent delay

Move new hits earlier - milliseconds compensates while recording; a positive value places new hits earlier. This is different from **Move take**, where a positive value moves an already recorded take later. Make a short test and listen before recording the full song.

> ### The optional beat guide is not a tempo requirement
>
> Turn on Show optional beat guide only when you know the tempo and first-beat time. It draws beats, not measures, and does not move your strikes. **Snap chosen take to eighth beats** changes timing only when you press that button; use Undo if it removes a feel you wanted.

With the recording canvas focused: Delete or Backspace removes the selected hit; Left/Right moves it 10 ms, or 100 ms with Shift. Save project retains the recording events and backing audio.

Import a MIDI file

Turn existing pitched MIDI parts into student strikes in the section-based editor.

1. Enable the Boomwhacker octaves you want available. Then open **Sound > Import MIDI** and choose a .mid or .midi file.
2. Select the tracks containing the student parts. Drum-channel notes are excluded.
3. Choose **Selected section** to replace its notes and meter, or **New sections, added to song** to create new material.
4. Decide whether to fold notes outside the enabled tube range into that range. Without folding, out-of-range pitches can be omitted.
5. Import, then inspect the pitches, chords, rhythms, section lengths, and full-song order before performing.

Imported information	Behavior
Note starts	Become strikes. Import is not a full notation-layout or instrument-orchestration system.
Tempo changes	Stay relative to Song tempo and repeat with their section, using quarter-note tempo units.
Time signatures	2/4, 3/4, 4/4, and 6/8 are supported, with changes on measure boundaries.
Existing lyrics	Replacing a section keeps its lyric timing, which may need adjustment to match the new music.

Use new sections for mixed meter

Choose New sections, added to song for a MIDI file that changes time signature. Replacing a reused section changes its notes and meter everywhere that section appears.

File import versus live recording

Import MIDI reads an existing file into musical positions. Connect MIDI keyboard in Record Along records new taps against audio time. Use the former to begin from an already sequenced composition, and the latter to perform a part directly with a recording.

After an import, simplify dense material for your class. For Bounce Along, eighth notes or slower and modest chord ranges are easier to keep readable than a dense MIDI arrangement.

Instrument introductions

Show students what to collect and where to play before the count-in begins.

1. In Stage, turn on **Show instrument introduction**.
2. Choose **One group at a time** or **All together**. Only instruments used in the playback scope appear.
3. Choose an Intro background and hold time. Use **Preview introduction** to watch it silently, with Pause and Previous/Next group controls.
4. For barred instruments, enable **Find your bars before playing** and optionally Include bar-removal tips. Check the separate bar preview as needed.

Presentation	Timing and organization
One group at a time	Low notes, regular tubes, high sample pitches, unpitched percussion, and each barred instrument appear as separate groups. Percussion splits into groups of up to six.
Hold each group after arrival	3, 5, 8, or 12 seconds. Bar diagrams remain visible for at least 6 seconds.
All together	Combined instrument screen held for 5, 8, 12, or 16 seconds, with separate bar-preview pages when enabled.
Find your bars pages	Full racks with needed positions highlighted; All together shows up to two instruments per page for 6 seconds.
Removal-only setup screen	When shown separately without Find your bars, the setup hold options are 6, 10, 15, or 20 seconds.

Instruments arrive with a fly-in, hold, then fade away in place. Intro and stage scenery are separate choices. After the introduction, the stage gets a two-second preparation period before the count-in. The opening avoids dropping students straight onto the first strike.

An introduction that matches the equipment

Choose the bass equipment under Instruments first. Octavator captions explain the tube and sounding pitch. Bar diagrams show physical positions, and full instrument names stay visible during the song. Preview again after changing the arrangement.

Intro background choices

Seventeen backgrounds for the instrument introduction. These do not replace the performance scene.

Family	Available backgrounds
Simple / gradients	Black; Midnight Gradient; Sunset Gradient.
Wood / classroom surfaces	Warm Wood; Herringbone Parquet; Acoustic Studio Panels.
Stone / architecture	Brick Wall; Desert Sandstone; Dark Slate; Sunlit Adobe.
Metals	Brushed Metal; Copper Patina; Hammered Bronze.
Fabric	Theater Velvet.
Animated light / weather	Underwater Light; Rainy Window; Leaf Shadows.

Choose and preview

Open **Stage > Intro background** or **Browse backgrounds...**. Instruments and lettering retain their soft drop shadows, and lettering adapts to the background. Use Preview introduction to judge the result with your actual instrument list.

Adjust the motion

Underwater Light has moving caustics, Rainy Window has sliding droplets, and Leaf Shadows gently sways. **Background motion** offers Still, Slow, Gentle, or Lively. **Light intensity** controls the light/effect strength. The other background choices stay still.

> ### If the background is not moving
>
> First check that you chose one of the three animated intro backgrounds and that Background motion is not Still. A device or browser reduced-motion preference keeps these backgrounds still.

Choose for the lesson

- **Instrument recognition:** start with Black or Midnight Gradient to make colors and shapes stand out.
- **Warm acoustic ensemble:** try Warm Wood, Sunlit Adobe, or Theater Velvet.
- **A calm opening:** try gentle Underwater Light or Leaf Shadows with a lower intensity.

Older saved intro choices map to the closest supported appearance in this version.

Classroom performance

Choose the display that suits the class, then rehearse the entrance and the transitions.

View	Best fit
3D Stage	Perspective falling notes, classroom instruments, and illustrated scenery.
2D Play Field	Straight falling-note lanes with an independent style and background.
Bounce Along - Boomwhackers	Two rows of readable tube notes followed by a bouncing ball; tubes only.

Transport and rehearsal

Choose Full song or Selected section, then Play. Use Pause, Restart, or the position slider as needed. **Loop rehearsal section** repeats a measure range in Full song; in section preview it loops the selected section. Loops repeat without another count-in.

Stage offers a **Silent lead-in**, **1 measure**, or **2 measures** for the count-in. Look ahead offers 2, 4, or 8 beats in section-based falling-note playback. These settings help students anticipate the entrance.

Clear the screen

- **Perform** hides the editing workspace. Full screen expands the display; Escape leaves performance mode.
- **Hide song info** above the preview, or Show song info in Stage, toggles the title, branding, tempo, section, and measure information.
- **Pitch names on bars, targets, and circles** and **pitch names on moving notes** are separate switches. Moving-note names are off by default.
- Hiding song info leaves lyrics, count-ins, previews, and instrument names available.

> ### Timing on your classroom system
>
> Visual timing adjustment runs from -500 to +500 milliseconds. Positive values make the visual cue arrive earlier. Use a short passage to check your projector/speaker setup. This does not retime the imported audio or repair an incorrect tempo map.

For the ending, choose the default, a custom message with an optional second line, or No message. Save to keep these choices.

3D stage scenes

Choose Stage background while using 3D Stage. The instrument intro has its own background selector.

Scene family	Current choices
Stage & performance	Midnight 3D stage; Deep space runway; Concert Lights; Concert Hall / Auditorium.
Mystery & adventure	Moonlit Graveyard; Spooky Witch House; Backrooms; Pirate Ship.
Landscape & atmosphere	Lava Cavern; Crystal Ice; Underwater Glow; Aurora Ribbon.
Classroom & countryside	Farm; Music Classroom; Autumn Pumpkin Patch; Storybook Christmas.
Seasons	Spring - Rain & Wildflowers; Fall - Golden Woodland; Winter - Snowfall; Summer - Sunlit Meadow.
Holidays	Hanukkah - Candlelight; St. Patrick’s Day - Emerald Hills.
Places & celebrations	American - Stars & Stripes; Canadian - Maple Lakeside; Mexico - Festival Courtyard; International - Flags Around the World; Albuquerque - Balloon Fiesta.

Movement supports the scene

The scene family includes seasonal motion such as rain and flowers, falling leaves, snow, and summer light with birds. Holiday and country scenes use their own decorative movement; Balloon Fiesta features balloons by the Sandias. Storybook Christmas combines fire, tree lights, and snow outside.

The backdrop stays behind the play area. A scene change does not change tube pitch colors, the instrument mix, or note timing. Animated scenery can continue during preparation rather than waiting for the first musical beat.

Check the actual classroom view

Try your chosen scene in Perform and watch a dense passage. Keep the musical cues easy to separate from the scenery. If the computer struggles, try a simpler stage or a still 2D background and reduce 2D effects when applicable.

These are the current selectable scenes, not a list of every background used in earlier versions. Removed themes and proposed future themes are not included in this guide.

2D styles & backgrounds

Style changes the playfield treatment. Background changes the scenery behind it.

2D style	Character
Original	The original falling-note aesthetic.
Classic Boomwhackers	Cylindrical tube targets with visible openings and impact feedback.
Laser Rush	Glowing arcade treatment and bright strike feedback.
Retro Music Studio	A music-studio visual treatment.
Candy Shop	A colorful candy-inspired playfield.

Choose the background independently

2D background offers Dark, Midnight Bokeh, Deep Water, Liquid Ink, Velvet Stage, Sunset Fade, Frosted Glass, Classic Blue, Warm Paper, and Chalkboard. Switching backgrounds does not switch the 2D style.

Motion and feedback

Animate background controls background movement where available. **Effects** offers Subtle or Full. Choose enough impact feedback to make the strike moment clear without distracting students from the next note.

In the falling-note views, play when the note reaches its target/strike point. With Classic Boomwhackers, the opening and impact flash support the illusion of a note entering the tube. Watch the arrival at the target rather than waiting for the whole note to disappear.

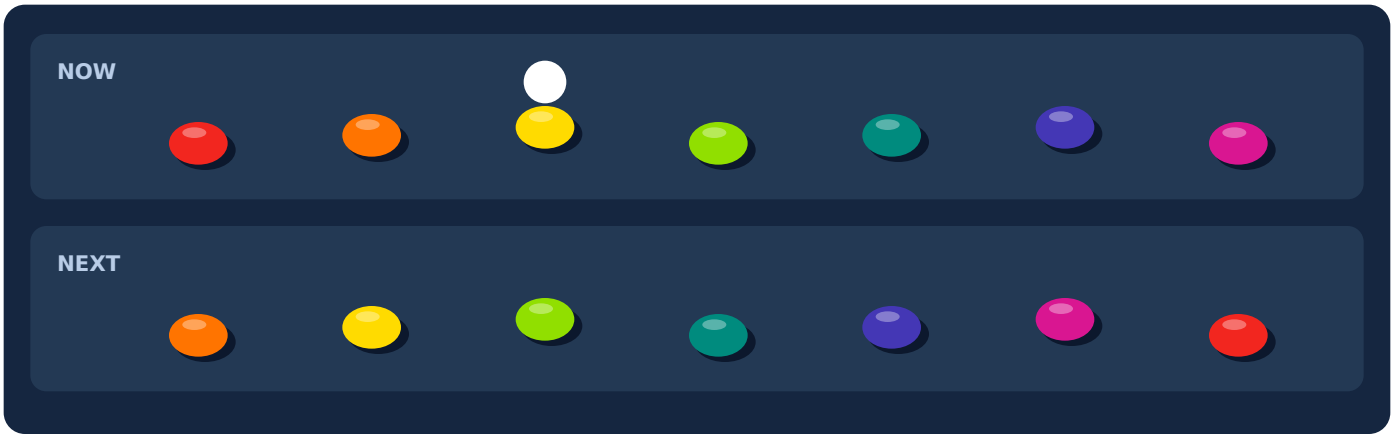
> ### The three visual selectors are different
>
> **Intro background** affects the instrument preparation screens. **2D style + 2D background** affect the flat playfield. **Stage background** affects the 3D scene. The app remembers these choices so switching views does not require rebuilding the look.

Instrument support

2D supports Boomwhackers, mallet parts, percussion, body percussion, lyrics, and Record Along strikes where those modes provide them. Barred instruments keep their material-colored bars and instrument names. Their falling pitch cues retain the pitch colors.

Bounce Along

Students read left to right while a ball lands on the next musical event.



Schematic: both rows remain available so students can read ahead.

The two rows alternate continuously. While the lower row plays, the completed top row fades and refreshes with the next phrase; the other row then follows. At a row change, the ball exits and re-enters from the side rather than traveling diagonally across the board. It glides through rests without a strike impact.

Control / behavior	What to expect
Adapt measure spacing	Two rows remain visible; crowded rhythms can use shorter rows to preserve note size.
Two measures each	Targets four measures on screen. If the rhythm or chord height cannot fit, the app asks you to simplify rather than shrink the notes excessively.
Show row labels	On by default. Toggles Now, Next, Played, measure captions, and arrows during classroom playback.
Notes and chords	Rounded, shaded notes with close shadows; pitch spacing is consistent. Chord pitches stack vertically; eligible paired eighths share beams.

Styles: Clean Midnight - Original, Wooden Music Toy, Storybook Meadow, Desert Sunset, Paper Cutout Theater, Moonlit Treehouse, and Underwater Pearl Garden.

Use melody-led patterns, eighth notes or slower, and modest chord ranges for the clearest view. Other instrument parts and the separate singing melody are not played here; bass parts assigned to barred instruments are excluded. Record Along uses 2D or 3D instead.

Troubleshooting & shortcuts

Check the small setting that explains the symptom before rebuilding your arrangement.

Symptom	First checks
No sound or a quiet part	Start audio with Play or Hear. Check the device output, guide checkbox, enabled pitch/part, Mixer level, and overall percussion/backing volume.
Part missing from the board	Check No Boomwhackers, octave switches, instrument enable and pitch choices. Bounce Along shows only supported tube parts.
Instrument missing from the intro	It must have used notes in the current playback scope. Enable the introduction and check bass equipment / instrument selection.
Cannot lift a bar off	Choose bars myself; NEEDED bars are protected. Enable bar-removal tips if you want the choices shown to students.
Lyrics missing or too late	Turn on Show lyrics, assign cue times, and increase Show words lead. Test section-boundary lyrics in Full song.
Audio drifts out of sync	Check BPM and the first downbeat, or use Align to audio. For free timing, use Record Along.
MIDI keys do nothing	Connect MIDI, allow access, choose the input, and check note mapping/range. Close dialogs and leave text fields before using computer keys.
Bounce Along will not start	Read its layout warning. Reduce chord height/range, use slower rhythms, choose adaptable spacing, or hide lyrics if they take needed room.
Changes did not save	Use Save project and keep the downloaded file. Open that saved copy next time; the app does not automatically save your arrangement.

Workspace shortcut	Action
Space / R / Escape	Play-pause / restart / leave performance.
Ctrl or Command + Z	Undo. Add Shift to redo.
Arrow keys on a lyric marker	Nudge it by the chosen Lyric timing step.

Shortcuts respect text fields and dialogs. Space on a focused button activates that button. In timing modes, Space is the tap command. Use the visible controls when keyboard focus is unclear. The ? button opens the app’s built-in help.

Classroom recipes & reference

Start with a simple musical goal, then choose the controls that serve it.

Goal	A practical setup
Body-percussion play-along	No Boomwhackers; enable Stamp, Pat, Clap, and Snap. Compose sections, or use Record Along for exact taps with a song. Choose 2D or 3D.
Pentatonic Orff ensemble	Enable the appropriate xylophone/metallophone parts. Write the music, choose C/A-minor or F/D-minor setup tips, and preview Find your bars.
Singing with accompaniment	Keep the accompaniment on the board. Add an audio-only Doo or flute melody, time the lyric phrases, and let words appear early.
A clear tube-reading activity	Use Bounce Along with eighth notes or slower, a modest pitch range, and two visible rows. Leave row labels on for the first rehearsal.
A focused rehearsal	Preview one section, or set a loop for a tricky passage. Lower guide sounds as students gain confidence, then play the whole song.

Quick control locator

Hide title: above the preview or Stage.
No tubes: Instruments > tube set.
Remove bars: instrument's Set up bars, or Stage > Choose bars.
Quieter guide, same cues: Sound > Mixer.
Earlier lyrics: Lyrics > Show words.
Changing recording tempo: Sound > Align to audio.
No known tempo: Create with > Record Along.
Now/Next labels: Stage > Bounce Along > Show row labels.

Open the app

Boomwhacker Play-Along - Teacher Edition

Created by Drew Wichman, Mr. Drew's Music Lab. This Teacher Edition guide covers composing, instrument setup, lyrics, audio, and live classroom performance.

Teacher Edition guide updated October 9, 2026. Examples and diagrams are explanatory illustrations. Available choices may change in later updates.

Enjoying Drew's free classroom tools? Buy Me a Coffee